

# SENTINEL-AI: FORENSIC INTELLIGENCE REPORT

Lead Analyst: Shalom Kumar | Session: 2025-26 | Version: 4.5.0

Institution: ISBM University | Faculty of Computer Applications

Advanced Pattern Recognition & Heuristic Weightage Security Audit

## SECTION 1: Target Ingestion & Identification

Target Telemetry Stream: `http://localhost/sentinel/public/bomb.php?payload=recursive_loop_infinite_exhaustion`

Forensic Audit ID: SENT-E00C7921

Analysis Timestamp: 2026-05-25 | 17:22:24

## SECTION 2: Intelligence-Based Risk Metrics

Heuristic Risk Index: 0/100

Security Verdict: SAFE / CLEAN

## SECTION 3: Detailed Forensic Logic Steps

The Sentinel-AI engine executed a weighted heuristic audit. The following threat signatures and behavioral anomalies were identified within the telemetry packet:

[+] CLEAN\_NODE: No immediate behavioral anomalies detected.

## SECTION 3.5: Statutory Cyber Law Mapping Analysis Matrix

### STEP 1: PERIMETER INGESTION & VOLUMETRIC TRAFFIC MONITORING

Technical Context Captured: IP Gateway Evaluation - Volumetric Traffic Clearance Passed [Weight: 15%]

#### LAW COMPLIANCE BOUNDARY: Section 43(f) of the Information Technology (IT) Act, 2000

Statutory Provision: Prevention of Denial of Service (DoS) and Unauthorized System Resource Depletion.

Forensic Verification Audit: The boundary subsystem calculated the incoming transaction frequency profile of the packet stream. Rate limiting thresholds successfully prevented infrastructure thread starvation.

### STEP 2: HEURISTIC PATTERN SCRAMBLING & DEEP SYNTAX ANALYSIS

Technical Context Captured: RegEx Logic Node Sweep Over Telemetry Token Components [Weight: 0%]

#### LAW COMPLIANCE BOUNDARY: Section 66 Read With Section 43(i) of the IT Act, 2000

Statutory Provision: Computer Related Offences - Tampering, Destroying, or Altering Destructive Source Code Assets.

Forensic Verification Audit: Heuristic inspection matrix executed tokenized pattern evaluation loops. Parallel processing nodes scanned configuration streams for unauthorized system administration calls.

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## STEP 3: NETWORK TRAJECTORY TRACING & REDIRECT LOOP ISOLATION

Technical Context Captured: Layer 3 Sniffer Loop Interception - Trajectory Routing Status [Weight: 0%]

**LAW COMPLIANCE BOUNDARY: Section 43(f) & Section 66F of the Information Technology Act, 2000**

Statutory Provision: Protection Against Cyber Terrorism and Attempted Infrastructure Lockout via Resource Loops.

Forensic Verification Audit: Active look-ahead sniffer framework intercepted remote host response headers. Redirection ceiling constraints successfully killed recursive redirect targets (Hmax Limit: 3).

## STEP 4: CRYPTOGRAPHIC EVIDENCE SEALING & IMMUTABILITY RECONCILIATION

Technical Context Captured: SHA-256 System Non-Repudiation Structural Signature Block [Weight: 0%]

**LAW COMPLIANCE BOUNDARY: Section 65B of Indian Evidence Act & Section 66C of IT Act, 2000**

Statutory Provision: Admissibility of Electronic Records and Protection of Identity Integrity Verification Metrics.

Forensic Verification Audit: System combined raw telemetry inputs, server session keys, and risk indicators into an immutable string hash block. Localized SHA-256 calculation sealed the digital evidence structure.

## SECTION 4: Forensic Integrity Verification

**SHA-256 DIGITAL FINGERPRINT (NON-REPUDIATION) :**

f57ef89c08b5659777349e1ae4cacc740eeaa4ce2864620a58f261b34bd5bf71

**EXECUTIVE SUMMARY:** Sentinel-AI has concluded the audit for the specified telemetry. The resulting 0% risk index warrants ROUTINE MONITORING. Data persisted to RDBMS logs under cryptographic law-compliance verification frameworks.